DSC 180A · SECTION D06 · WEEK 5 CHECK-IN

Differentiable Climate Emulation in JAX

From ClimateBench emulators to a climate model we run ourselves

TEAM

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MENTOR

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PROJECT OVERVIEW

Climate models are too slow to test every emission pathway

Inputs

CO_2 and CH_4 (global), SO_2 and black carbon (maps)

Emulator

Learns the map from inputs to climate. Runs in seconds.

Outputs

NorESM2 maps of tas, DTR, pr and pr90

ClimateBench: train on 6 NorESM2 scenarios, test on ssp245, score with NRMSE.

Our angle: build the emulator in JAX so it is differentiable, and learn the physics by running a real atmosphere model (JCM).

PROJECT GOALS · PHASE I

Three steps to a differentiable emulator

Weeks 1–3

Reproduce ClimateBench

Four baselines, our own code, scored against the published leaderboard

Done

Weeks 4–7

Run a climate model

JCM: forcing, feedback and climate sensitivity, then jax.grad through the model

In progress

Weeks 8–10

Bring them together

A ClimateBench emulator in JAX, gradients for scenario analysis, a Phase II proposal

Next

METHODOLOGY

Four emulators and one climate model

Emulators · Weeks 1–3

- Pattern scaling with true or EBM-predicted global T, also ported to JAX
- Gaussian process with a linear CO_2 kernel
- Random forest on CO_2, CH_4 and aerosol EOFs
- CNN-LSTM
- One shared NRMSE module, leave-one-scenario-out CV, a leakage audit

Climate model · Weeks 4–5

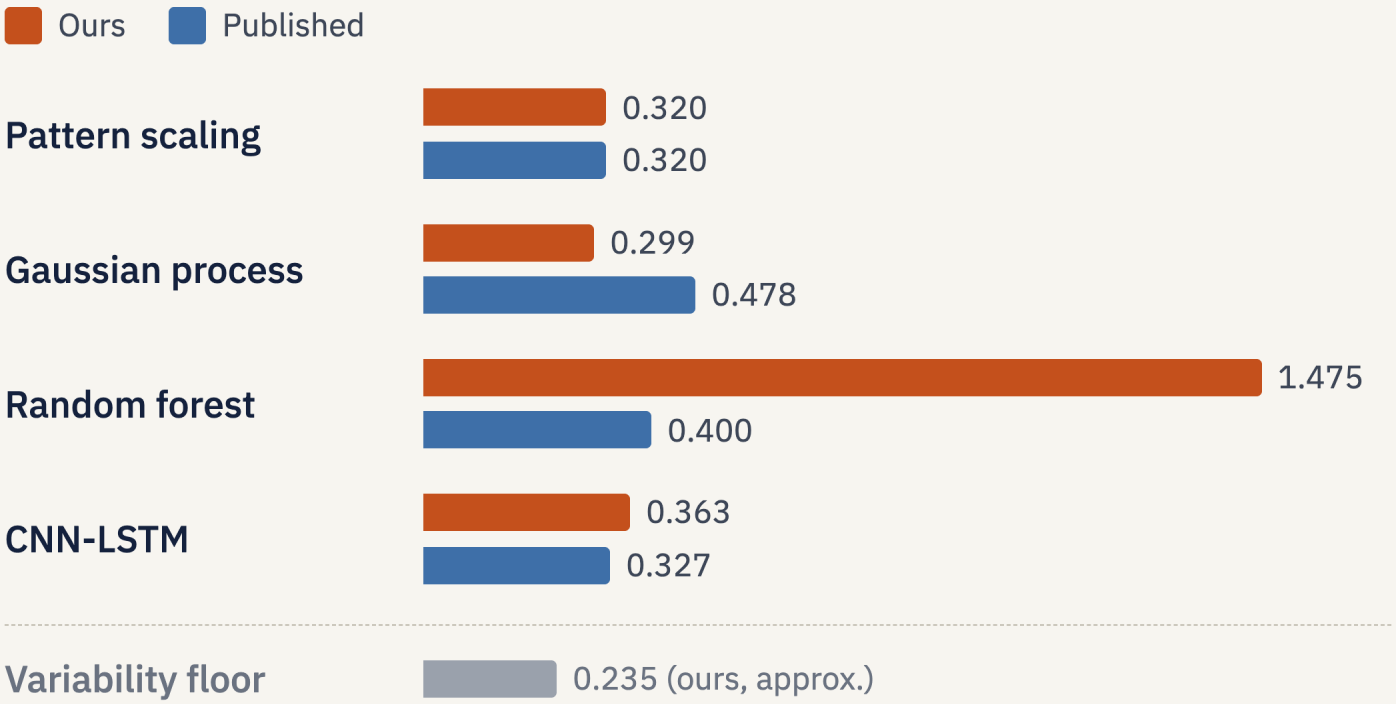
- JCM SPEEDY: T31 grid (96 × 48), 8 levels, 15-min step
- Fixed SSTs: ERF = change in TOA net flux
- Slab ocean (JEM): feedback λ and ECS
- NRP GPU (GTX 1080 Ti), because the DataHub GPU does not work

NOTES.md decision log: what we asked the AI, what we changed, and what independent evidence we checked it against

RESULTS · WEEKS 1–3

Leaderboard: 3 of 4 baselines reproduced

Total NRMSE for tas, ssp245, 2080–2100 (lower is better)



Reproduced: pattern scaling (exact), GP and CNN-LSTM

GP beats the paper on tas because of our linear CO_2 kernel

Random forest misses: its warming levels off at ~1.4 K, 0.5 K too cold

Best models are 1.3–1.9× the variability floor

Source: results/leaderboard.csv · Group D06

CHALLENGES

Most of the problems were compute, not science

The DataHub GPU does not work	We moved to the NRP JupyterHub (GTX 1080 Ti)
NRP limits: 5 GB home, 1 CPU core, stops 1 h after the browser closes	Environment in /tmp, runs under nohup, output cut to 17 MB
“pip install jcm jax-esm” fails	One script installs both at pinned commits
The AI read CO_2 = 360 ppmv from an old code comment	We loaded the forcing and found 420 ppmv; we check values in code and data

TEAM CONTRIBUTIONS

Each member owned one emulator

Annaly Ellis

- Pattern scaling with true and EBM-predicted global T
- Port to pure JAX (lstsq, jax.grad + vmap), equal to numpy to 1e-15

Coltrane Gowan

- Random forest with a CV hyperparameter search
- Week 3 answers and presentation
- NRP setup, throughput and control run (Week 4)

Linus Lee

- CNN-LSTM emulator and its result files
- Figures and the Week 3 presentation notebook

Roman Respicio

- Repo, environments and the decision log
- Week 1 maps and time series
- Gaussian process, shared metrics, leaderboard

NEXT STEPS

Next: let the ocean warm

Week 5 · slab ocean (JEM)

Feedback and climate sensitivity

- Q-flux slab control for a few decades; check the SST drift
- Abrupt 2× and 4×CO_2 runs to equilibrium
- Gregory regression: ERF and λ, compared with our fixed-SST ERF
- ECS vs. NorESM2-LM (~2.5 K) and the IPCC AR6 range
- ΔT and ΔP patterns vs. our Week 3 pattern scaling

Week 6

jax.grad of ERF and warming vs. SPEEDY parameters, checked by finite differences; a twin experiment

Week 7

Perturbed-parameter ensemble; a GP emulator of ERF, λ and ECS

Weeks 8–10

Best baseline in JAX, gradients w.r.t. emissions, Phase II proposal